05 • DATABASE SCHEMA

Database Schema

Where the information is kept, how the pieces relate, and who may see and change what.

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Planning documents for the Sali app

GitHub: supermarket-prices Live site: sali.one

IN ONE MINUTE

- Think of a spreadsheet with **tabs**: Chains, Stores, Products, Prices, Promotions, plus two work-log tabs.
- **Prices** link a product to a store. History is kept, but only one price per product and store is “current”.
- No personal data is kept at all. The public side can only **read**.

29	2,000+	240,000+	2×
chains	stores	products	refreshed daily

01 The tables

Chains `chains`

Name (Shufersal, Rami Levy…)
Official company ID

Stores `stores`

Which chain, branch number
Name, city, address
Map position and its accuracy
When prices and promotions were last confirmed

Products `products`

Barcode (one ID everywhere)
Name, brand, unit
Picture

Prices `prices`

Which product, at which store
The price, exact to the agora
Current or history

Promotions `promotions`

The chain's promotion text
One row per product it covers
Quantity, price, dates
Whether it can be applied in a basket

Work logs `ingest_*`

Every file loaded and its state
Every refresh run, per chain

A chain has many stores • a store has many prices • a product is priced at many stores • a price = product + store + amount

Two pre-worked tables are refreshed after every run: per product its store count and lowest and highest price, and per promotion its price per unit and saving against the cheapest shelf price. Search ordering, the home-page examples and Deals read from them, so they are fast.

02 The data rules that matter

Each is enforced in the code and by the database itself.

Rule	How it works	What it prevents
One current price per product per store	A changed price marks the old row as history and adds a new one; the database refuses two current rows.	Two prices at one store, or lost history
Prices exact to the agora	Stored as exact decimals, never floating-point numbers.	Totals like 9.7999
A store is chain + branch number	Branch "001" exists in many chains, so the pair is the identity and is unique.	One chain's prices landing in another's store
The first product name stays	If another chain names the product differently, the stored name is kept.	Names flipping between refreshes
A file is "loaded" only once its data is saved	The log is written in the same step as the data; a file that fails three times is put aside.	Double loads, or "loaded" files with no data

Why unchanged prices are not rewritten. About 11 million prices are checked daily and most do not change, so only one field on the store (“last confirmed”) is updated. Rewriting every row kept the database saturated for hours.

03 Who may see and change what

No personal data is held, but permissions still matter.

Who	Allowed	Not allowed
A visitor	Search, compare, build a basket. No login.	No direct database access
The API (read-only user)	Read every table; set a product’s picture link.	Change prices or names, delete, alter tables (tested)
The Worker (owner)	Load data and change structure.	Not reachable from the internet

Baskets and chosen filters live only in the user’s browser. The user’s location is never kept: a rounded point may be sent once to suggest a city, then discarded. Every structure change is one of 20 recorded, reversible migrations, and nothing in price history is deleted.